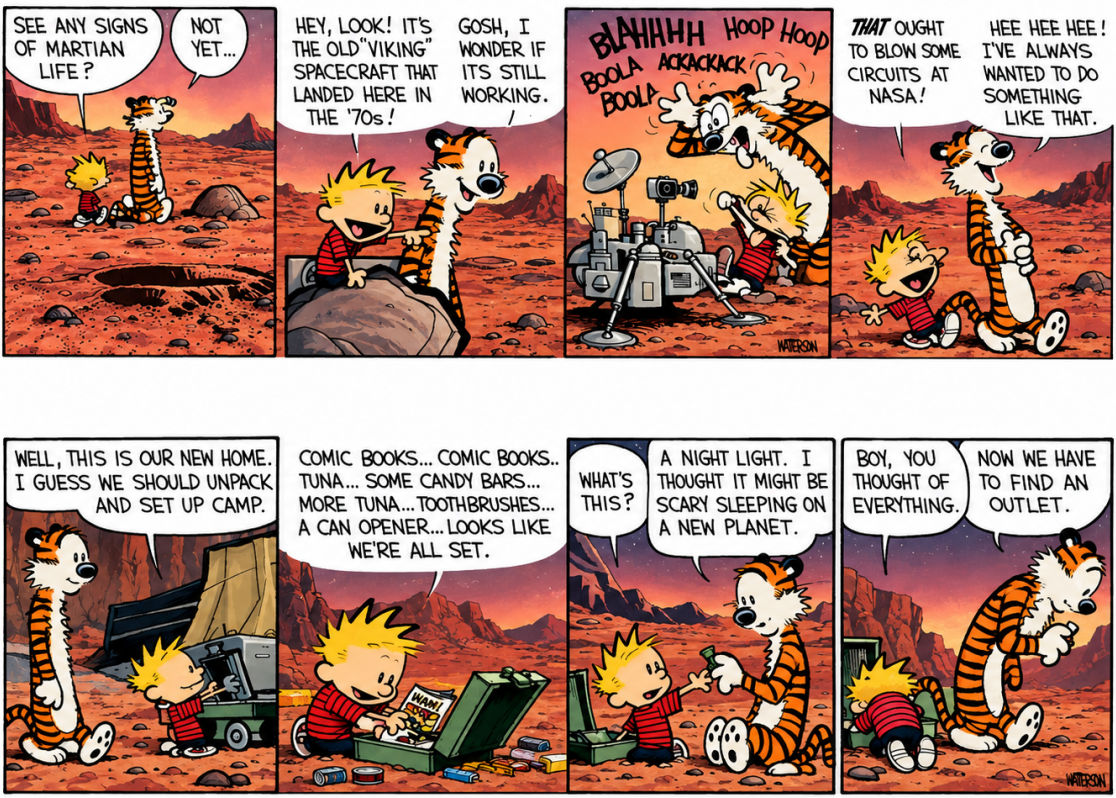
Session - 15

Sagemaker canvas and feature store 2

Nov 04, 2026



	customer_id	event_time	orders_so_far	returns_so_far	return_rate	tenure_days
0	C-00001	2025-10-13T00:00:00Z	1	0	0.1667	790
1	C-00001	2025-10-17T00:00:00Z	2	0	0.1429	794
2	C-00001	2025-10-27T00:00:00Z	2	1	0.2857	804
3	C-00001	2025-12-07T00:00:00Z	3	1	0.2500	845
4	C-00001	2025-12-21T00:00:00Z	3	2	0.3750	859
5	C-00001	2025-12-22T00:00:00Z	4	2	0.3333	860
6	C-00001	2026-01-05T00:00:00Z	4	3	0.4444	874
7	C-00001	2026-03-01T00:00:00Z	5	3	0.4000	929
8	C-00001	2026-03-15T00:00:00Z	5	4	0.5000	943
9	C-00001	2026-04-04T00:00:00Z	6	4	0.4545	963

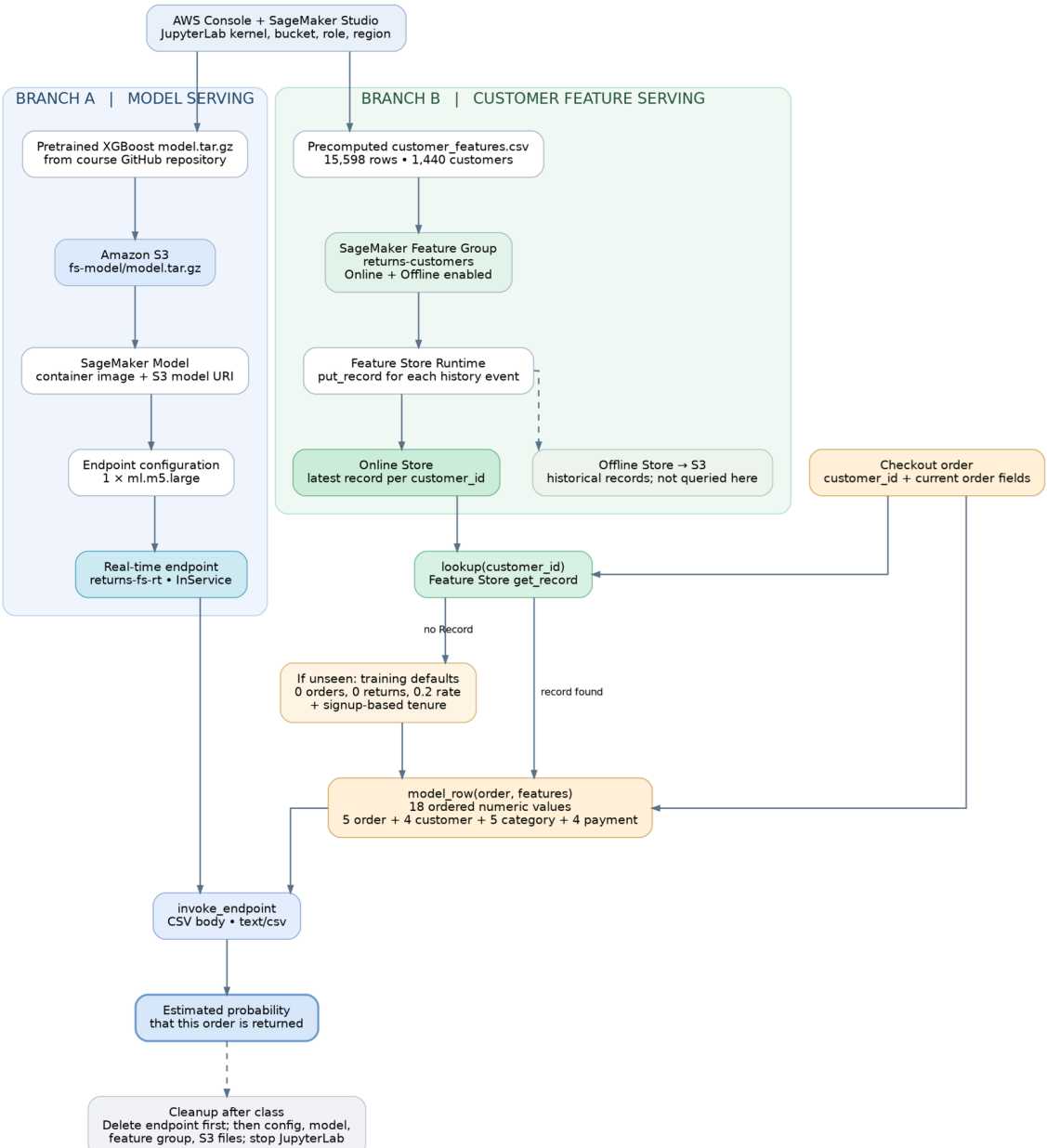
	order_id	customer_id	order_date	category	price	discount_pct	delivery_days	payment_method	is_gift	seller_rating	returned	signup_date
402	ORD-200402	C-00001	2025-10-13	apparel	1481.0	18	3	card	0	4.0	1	2023-08-15
528	ORD-200528	C-00001	2025-10-17	home	3164.0	37	2	cod	0	3.9	0	2023-08-15
2200	ORD-202200	C-00001	2025-12-07	apparel	684.0	16	1	upi	0	4.5	1	2023-08-15
2684	ORD-202684	C-00001	2025-12-22	books	1179.0	42	9	upi	0	3.8	1	2023-08-15

order_id	customer_id	order_date	price
O1	C-01	2026-01-10	500
O2	C-01	2026-01-24	800
O3	C-01	2026-02-07	600
O4	C-01	2026-02-21	900

customer_id	event_time	orders_so_far	returns_so_far
C-01	2026-01-10	1	0
C-01	2026-01-24	2	1
C-01	2026-02-07	3	2
C-01	2026-02-21	4	3

Order being predicted	Eligible snapshot dates	Selected snapshot	Historical orders	Historical returns
O1 — 10 Jan	None	**No match**	—	—
O2 — 24 Jan	10 Jan	**10 Jan**	1	0
O3 — 7 Feb	10 Jan, 24 Jan	**24 Jan**	2	1
O4 — 21 Feb	10 Jan, 24 Jan, 7 Feb	**7 Feb**	3	2

SageMaker Feature Store — Part 2: Live Predictions



```python
def lookup(customer_id):
    answer = store.get_record(FeatureGroupName=FEATURE_GROUP, RecordIdentifierValueAsString=customer_id)
    return {feature["FeatureName"]: feature["ValueAsString"] for feature in answer.get("Record", [])}


def model_row(order, features):
    if not features:   # never seen: no orders, no returns, tenure from the signup date
        days = (pd.Timestamp(order["order_date"]) - pd.Timestamp(order["signup_date"])).days
        features = {**NO_HISTORY, "tenure_days": days}
    return ([order[c] for c in ORDER_COLS]
            + [float(features[c]) for c in FEATS]
            + [int(order["category"] == c) for c in CATEGORIES]
            + [int(order["payment_method"] == p) for p in PAYMENTS])


def predict(order):
    row = ",".join(str(value) for value in model_row(order, lookup(order["customer_id"])))
    answer = runtime.invoke_endpoint(EndpointName=ENDPOINT, ContentType="text/csv", Body=row)
    return float(answer["Body"].read()), row
```

Features From online feature store

